Identifying the Limiting Reactant

PhET Reactants, Products and Leftovers · PhET

Senior Secondary (Years 11-12)	~30 minutes	Science · Chemistry

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Simulation: https://phet.colorado.edu/en/simulations/reactants-products-and-leftovers
Curriculum links: chemical reactions · stoichiometry

Learning intentions

- I can apply the sandwich-making analogy to identify a limiting reactant in a chemical reaction.
- I can use mole ratios from a balanced equation to predict the amount of product formed.
- I can identify which reactant is left over and calculate the excess amount.

Getting started

Open the simulation and start on the Sandwiches screen. Set a 'recipe' (e.g. 1 bread + 1 cheese + 1 ham = 1 sandwich) and choose starting amounts of each ingredient before moving to the Chemistry screen for the later tasks.

Tasks

1. On the Sandwiches screen, set unequal amounts of bread and cheese (e.g. 4 bread slices, 6 cheese slices for a 1:1 recipe). Which ingredient runs out first, and how many sandwiches can you make?

2. Name the ingredient that is 'left over' in the previous task, and state how much of it remains.

3. Explain, using your sandwich example, what a 'limiting reactant' would mean in a real chemical reaction.

4. Move to the Chemistry screen and select a reaction. Record the balanced equation, then use the given starting amounts of each reactant to calculate, by hand, which reactant is limiting.

Working:

5. Using your answer to the previous task, calculate the theoretical amount of product formed, then check it against what the simulation shows.

Working:

6. Change the starting amount of the non-limiting reactant only. Does the amount of product formed change? Explain why or why not.

7. Now change the starting amount of the limiting reactant instead. Recalculate which reactant is limiting now (if it changes) and the new amount of product formed.

Working:

8. Explain why manufacturers need to consider limiting reactants when planning a large-scale chemical process, using cost or waste as part of your reasoning.

Extension (optional)

Choose a reaction on the Chemistry screen with more than two reactants, or design one on paper, and determine the limiting reactant among three species given their starting amounts.